

A counterexample to the Hermite–Bergman endpoint Toeplitz conjecture

Literature-implied answer to the conjecture in arXiv:0907.2844

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Status: literature-implied full negative answer. For every complex dimension $n \geq 1$ and every Hermite time $\tau > 0$, there is a symbol satisfying the source’s condition $(*)$ whose Toeplitz operator has a bounded extension, although its critical Gaussian regularization is unbounded. The source conjecture is therefore false.

Source conjecture

Jotsaroop and Thangavelu define the Hermite–Bergman space $\mathcal{H}_\tau(\mathbb{C}^n)$ using

$$U_\tau(x, y) = 4^n (\sinh 4\tau)^{-n/2} \exp(\tanh(2\tau)|x|^2 - \coth(2\tau)|y|^2).$$

For a measurable g satisfying

$$gK_\tau(\cdot, w) \in L^2(U_\tau, dv) \quad \text{for every } w \in \mathbb{C}^n, \quad (*)$$

they define the densely defined Toeplitz operator $T_g = P_\tau M_g$. The remark on source PDF page 12, immediately after the boundedness theorem in Section 3.2, conjectures

$$T_g \text{ extends boundedly on } \mathcal{H}_\tau(\mathbb{C}^n) \iff g * q_{\sinh(4\tau)/8} \in L^\infty(\mathbb{C}^n), \quad (1)$$

where q_s is the Euclidean heat kernel on $\mathbb{C}^n \simeq \mathbb{R}^{2n}$ [1].

Supporting theorem

In Looi’s standard Fock normalization

$$d\mu(\zeta) = (2\pi)^{-n} e^{-|\zeta|^2/2}, dv(\zeta), \quad K(\zeta, a) = e^{\zeta \cdot \bar{a}/2},$$

Theorem 1.2 on supporting PDF page 3 constructs, for every $n \geq 1$, a measurable h such that

1. $hK(\cdot, a) \in L^2(d\mu)$ for every a ;
2. T_h extends to a bounded operator on the Fock space;
3. $h * q_{1/4}$ is unbounded.

The paper proves that the bounded extension agrees with the natural Toeplitz operator on the dense span of kernel vectors, so its boundedness notion is the one used in the source conjecture [2].

Exact normalization transport

Proposition 1. *For every $\tau > 0$, Looi's symbol transports to a counterexample to (1). Hence no part of the source conjecture remains open for general measurable symbols satisfying (*).*

Proof. Put

$$S = \sinh(4\tau), \quad c = \coth(4\tau), \quad \lambda = \sqrt{2/S}.$$

First remove the nonradial part of the Hermite–Bergman weight. For an entire function f , set

$$(M_\tau f)(z) = e^{-cz^2/2} f(z).$$

The elementary identities

$$\coth(4\tau) - \tanh(2\tau) = S^{-1}, \quad \coth(2\tau) - \coth(4\tau) = S^{-1}$$

give, pointwise for $z = x + iy$,

$$|e^{-cz^2/2}|^2 U_\tau(x, y) = 4^n S^{-n/2} e^{-|z|^2/S}.$$

Thus M_τ is a unitary identification (up to the harmless normalizing constant in the measure) from a radial Fock space $\mathcal{F}_S$ with Gaussian scale S onto $\mathcal{H}_\tau$. Moreover,

$$\langle g M_\tau f, M_\tau k \rangle_{\mathcal{H}_\tau} = \langle g f, k \rangle_{\mathcal{F}_S}. \quad (1)$$

Consequently the Toeplitz forms, their natural domains, and existence of a bounded extension are preserved with the same symbol g .

Now dilate $\mathcal{F}_S$ to Looi's Fock space. After multiplication by a scalar chosen to normalize the norm, $f(\zeta) \mapsto f(\lambda z)$ is unitary, because $|z|^2/S = |\lambda z|^2/2$. If h is Looi's symbol, define

$$g_\tau(z) = h(\lambda z).$$

Equation (1) and the change of variables $\zeta = \lambda z$ show that the Toeplitz operator with symbol g_τ is unitarily equivalent to T_h . The reproducing kernels transform by the same holomorphic multiplier and dilation (up to nonzero scalar factors), so Looi's condition $hK(\cdot, a) \in L^2(d\mu)$ is exactly the source condition (*) for g_τ .

Finally, heat flow commutes with dilation according to

$$(g_\tau * q_s)(z) = (h * q_{\lambda^2 s})(\lambda z).$$

At the source endpoint $s = S/8$ one has $\lambda^2 s = (2/S)(S/8) = 1/4$. Therefore

$$(g_\tau * q_{S/8})(z) = (h * q_{1/4})(\lambda z),$$

which is unbounded by Looi's theorem, whereas T_{g_τ} has a bounded extension. This violates the forward implication in (1). $\square$

Provenance and scope

Looi explicitly disproves the standard Berger–Coburn conjecture, but the supporting paper does not cite arXiv:0907.2844 or state its Hermite–Bergman variant. The full answer above depends on the direct multiplier-and-dilation identification, so it is classified as a literature-implied answer rather than an explicit literature answer or a new counterexample. The conclusion is restricted to the source's general measurable-symbol conjecture; its positive theorems for structured symbols remain valid.

The run indexes were searched for the source id/title, Hermite–Bergman terms, the Berger–Coburn conjecture, and arXiv:2601.20859. They contained an existing packet for Looi's answer to the standard Fock conjecture, but no prior packet transporting it to this source conjecture.

References

- [1] K. Jotsaroop and S. Thangavelu, *Toeplitz operators with special symbols on Segal–Bargmann spaces*, arXiv:0907.2844 (2009).
- [2] S. Looi, *A counterexample to the Berger–Coburn conjecture*, arXiv:2601.20859 (2026).